

# LectureDoc2 Cheat Sheet

## Information

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**Version:** Work in Progress

## Admonitions

Standard reStructuredText admonitions: attention, caution, danger, error, hint, important, note, tip and warning

LectureDoc2 specific admonitions: background, definition, proof, theorem, lemma, conclusion, observation, remark, summary, legend, repetition, question, answer, remember, deprecated, assessment, and example.

|                                                                                                                                                         |                                                                                                                                                                                                                                      |                                                                                                                                                           |                                                                                                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>Generic Admonition<br/>with Custom Title</div> <div>This is a generic<br/>admonition.</div>                                                        | <div> Background</div> <div>This is a background<br/>admonition.</div>                                                                              | <div> Remark</div> <div>This is a remark<br/>admonition.</div>           | <div>Deprecated</div> <div>This is a deprecated<br/>admonition.</div>                                                                                                                                                                                                                   |
| <div> Attention!</div> <div>This is an attention<br/>admonition.</div>  | <div> Definition</div> <div>This is a definition<br/>admonition.</div>                                                                              | <div> Example</div> <div>This is an example<br/>admonition.</div>        | <div> Assessment</div> <div>This is an assessment<br/>admonition.</div>                                                                                                                              |
| <div>Caution!</div> <div>This is a caution<br/>admonition.</div>                                                                                        | <div> Definition: A<br/>Definition with a<br/>very long title that<br/>spans multiple lines</div> <div>This is a definition<br/>admonition.</div> | <div> Summary</div> <div>This is a summary<br/>admonition.</div>       | <h2>Support for Notes</h2> <p>Text above the note.</p> <p>Notes are<br/>floated to<br/>the right<br/>w.r.t. the content that follows<br/>the note.</p> <div> Note</div> <div>This is a note.</div> |
| <div> !DANGER!</div> <div>This is a danger<br/>admonition.</div>      | <div> Proof</div> <div>This is a proof<br/>admonition.</div>                                                                                      | <div> Legend</div> <div>This is a legend<br/>admonition.</div>         |                                                                                                                                                                                                                                                                                         |
| <div>Error</div> <div>This is an error<br/>admonition.</div>                                                                                            | <div> Theorem</div> <div>This is a theorem<br/>admonition.</div>                                                                                  | <div> Repetition</div> <div>This is a repetition<br/>admonition.</div> |                                                                                                                                                                                                                                                                                         |
| <div> Hint</div> <div>This is a hint admonition.</div>                | <div>Lemma</div> <div>This is a lemma<br/>admonition.</div>                                                                                                                                                                          | <div> Question</div> <div>This is a question<br/>admonition.</div>     | <div> Tip</div> <div>This tip is shown below the n<br/>because the class <code>clear-fl</code><br/>used to <i>clear the float</i> of the r</div>                                                   |
| <div> Important</div> <div>This is an important<br/>admonition.</div> | <div> Conclusion</div> <div>This is a conclusion<br/>admonition.</div>                                                                            | <div> Answer</div> <div>This is an answer<br/>admonition.</div>        | <div> Example</div> <div><pre>1 .. note:: 2     This is a note. 3 4 Text that is left of t 5 6 .. tip:: 7     :class: clear-float 8 9     This tip is shown .</pre></div>                          |
| <div> Tip</div> <div>This is a tip admonition.</div>                  | <div> Observation</div> <div>This is an observation<br/>admonition.</div>                                                                         | <div> Remember</div> <div>This is a remember<br/>admonition.</div>     |                                                                                                                                                                                                                                                                                         |
| <div> Warning</div> <div>This is a warning<br/>admonition.</div>      |                                                                                                                                                                                                                                      |                                                                                                                                                           |                                                                                                                                                                                                                                                                                         |

## Most basic Example

`.. definition::` A Definition with a very long title that spans multiple lines

This is a definition admonition.

## SVGs

SVGs can easily be embedded directly in RST documents. Direct embedding has the advantage that animation is then possible.

### External SVGs

*Usage: slide view and document view*

In general, SVGs should be designed with the slide view in mind. They will automatically be scaled in the document view based on the font-ratio between the base font size used in the slide view and the base font size used in the document view.

*Usage: document view only*

SVGs shown only in the document view (e.g. solutions to exercises) should be designed with the document view in mind and hence should use the base font size of the document view as the foundation.

### Embedded SVGs

They should be designed independent of any specific size and only use relative sizing. This ensures that they look beautiful in all contexts.

(See the respective example for more details.)

SVG

#### Example

```
1 .. raw:: html
2
3     <div style="width: 5ch; height: 5ch;">
4     <svg   viewBox="0 0 5 5"
5         font-size="2"
```

```

6         version="1.1"
7         xmlns="http://www.w3.org/2000/svg">
8         <rect x="0.5" y="0.5" width="4" height="4" fill="darkblue"/>
9         <text x="0.75" y="3" fill="white">SVG</text>
10    </svg>
11 </div>

```

## Embedded SVGs Only

### ▲ Attention!

|                |                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IDs:</b>    | A single SVG in an rst source file, will be found at least three times in the HTML: In the document view, the slide view, and the lighttable view. Hence, if an SVG uses <code>ids</code> , they are no longer unique which may cause rendering issues (in particular with Firefox). Therefore SVG elements with ids (typically markers) need to be defined using shared definitions. |
| <b>Styles:</b> | Note that all styles are global and affect all elements.                                                                                                                                                                                                                                                                                                                              |

Shared definitions in LD-rst documents:

### 📄 Example

```

1  .. meta::
2     <other meta information>
3     :svg-style:
4         g.graph {
5             circle{
6                 fill: var(--current-fg-color);
7                 stroke-width: 0.2;
8
9                 &.red { fill: red;}
10                &.green { fill: green; }
11                &.blue { fill: blue; }
12                &.start-node {
13                    fill: none;
14                    stroke: red;
15                    stroke-width: 0.3;
16                }
17            }
18        }
19     :svg-defs:
20         <marker
21             id="arrow"
22             viewBox="0 0 2 2"
23             refX="1.8"
24             refY="1"
25             markerUnits="strokeWidth"
26             markerWidth="6"
27             markerHeight="6"
28             orient="auto-start-reverse">
29             <path class="arrow-head" d="M 0 0 L 2 1 L 0 2 z" fill="context-stroke" />
30         </marker>

```

LectureDoc generally distinguishes between the slide view and the document view. Both views represent the content in similar ways, but the slide view is optimized for presentation and generally uses a much larger font size (e.g., 48px) while the document view uses a smaller font size (e.g., 14px or 16px). Hence, it is recommended to draw an SVG using a font-size of 1 (no unit!) and to put the SVG in a fixed sized container where the size of the container depends on the font-size.

### 📄 Example

SVG

## Vertical Titles

Add the class `vertical-title` to rotate the title and to change the layout of the slide to a column-based layout. To get back to a row based layout add a container with the class `width-100`.

```
.. class:: vertical-title
```

```
1 .. row:: html
2
3     <div style="width: 5ch; height: 5ch;">
4     <svg   viewBox="0 0 5 5"
5         font-size="1"
6         version="1.1"
7         xmlns="http://www.w3.org/2000/svg">
8         <rect x="0.5" y="0.5" width="4" height="4" fill="darkblue"/>
9         <text x="1.75" y="3" fill="white">SVG</text>
10    </svg>
11    </div>
```

<Slide Title>

```
.. container:: width-100
```

<Row based content layout.>

## Slide Tweaks

### Slide without Title

To hide the title of a slide assign the class `no-title`.

### Vertically Centered Content

Use the class `center-content`.

```
.. class:: center-content
        no-title
```

Hidden on the slide!

<Slide Content>

## Sections and Subsections

Create a slide that marks the beginning of a new section or subsection by adding the class `new-section` or `new-subsection` to the slide.

## Exercises and Solutions

Associate the class `exercises` with a slide to indicate that the slide contains an exercise.

Use the directive `.. exercise::` to add an exercise. To add a solution use the custom directive `.. solution::` inside of an exercise block and specify the password (optional) using the attribute `:pwd:`.

### Example (Solution in supplemental information)

```
.. exercise:: <Title>
```

<Description of the exercise.>

```
.. solution:: <Title>
:pwd: 1234
```

<Solution>

You can configure a master-password using meta information:

```
.. meta::
:master-password: <Master-password>
```

## Table of Contents

A navigable table of contents can be created using standard `rst` techniques.

## Example

## Table of Contents

---

- `Section 1 Title`\_  
`Subsection 1.1 Title`\_

## Footnotes

[#]\_ and .. [#] create footnotes.

Test\ [#]\_

---

.. [#] test.org

## Explicit Footers

A container with the class `footer-left`, `footer-right` or `block-footer`.

## References

Use standard rst references.

## Example

...  
Like described in [Eic24]\_ ...  
...

## References

---

.. [Eic24] LectureDoc2; 2024

## Copy to Clipboard

Add `copy-to-clipboard` to a code block to enable copying code to the clipboard.

## Example

```
.. code:: java
   :class: copy-to-clipboard

   public static void main(...)
```

## Fade-out Content

Add the class `fade-out` to a container to whiten the content.

## Supplemental Information

Use the directive `supplemental` for respective information.

## Example

```
.. supplemental::

   <Text>
```

## Text Alignment

Control text alignment: `text-align- [left | center | right]`

## Images

Adding a drop-shadow and rounded corners: `picture`.

## Tables

The layout can be adapted using: `compact`, `compact-cells`, `no-table-borders`, `no-inner-borders`, `no-column-borders`, `fake-header [-2nd] -row` and `fake-header [-2nd] -column`.

## Animation

`incremental` (and `wobble`).

`highlight-line-on-hover` (always usable), `highlight-on-hover` (explicit column or row headers are not supported) or `highlight-identical-cells`

## Lists

■ `list-with-explanations` renders text paragraphs of list items less pronounced.

(As shown here.)

■ Use `impressive` to make the list more impressive:

■ Add `negative-list` to use "❗" for bullet points.

■ Add `positive-list` to use "✓" for bullet points.

## Example

- Point 1

```
.. class:: negative-list list-with-explanations
```

- Point 2

Some on-slide explanation.

```
.. class:: positive-list
```

- Point 3

## Decorations

`line-above` draws a horizontal lines.

`box-shadow` adds a shadow.

`rounded-corners` the corners will be rounded.

## Example

```
.. container:: margin-top-1em
    line-above
    padding-top-1em
    box-shadow
```

Text

## Font Styling

**"rem" based relative sizes:** `xxl`, `huge`, `large`, `small`, `footnotesize`, `scriptsize`, `tiny`, `x-tiny`, `xx-tiny`

**"em" based relative sizes:** `larger`, `smaller`, `far-smaller`

**Font weight:** `bold`, `light`, `thin`

**Font family:** `monospaced`, `serif`

**Font style:** `italic`

## Slide Transitions

Available slide transitions: `transition-move-left`, `transition-scale`,  
`transition-fade`, `transition-move-to-top`

## Example

```
.. class:: transition-move-left
```

<Slide Title>

-----

## Revealing Slide Content

All elements with the class `incremental` are revealed incrementally.

## Example

```
.. class:: incremental
```

```
- Item 1 - Part 1
```

```
:incremental: Item 1 - Part 2
```

```
- Item 2
```

## Column-based Layouts

Use `two-columns` and `three-columns` for respective layouts.

### Example

```
.. container:: two-columns
```

```
.. container:: column no-separator
```

```
<Column 1>
```

```
.. container:: column
```

```
<Column 2>
```

Add `no-default-width` to the root container for content based column widths. Use class `no-separator` on the left column to remove the separator.

## Stacked Layouts

Stacked layouts are based on nested layers. Each layer - except of the first one - needs to have the class `incremental` and/or the class `overlay` for transparent layers. (Up to 10 layers are supported.) To turn off the numbering of opaque layers use `.no-number`.

## Images in Stacked Layouts

To avoid that a parent element of a floating element is collapsed add the class `clearfix` to the parent element; i.e., when a layer just contains a floating image.

### Example

```
.. deck::
```

```
.. card:: clearfix
```

```
.. image:: <p1.svg>  
:align: left
```

```
.. card:: overlay
```

```
.. image:: <p2.svg>
```

```
.. card:: warning
```

```
<Content>
```

## Semantic-based Text Markup

`peripheral`: for less important text. `obsolete`: for obsolete statements. `ger`: to markup German Words.

`eng`: to markup English words.

## Box sizes

Use `width-100%` and `width-75%` to control the width of a container.

## Colors (roles)

### Font Colors

red, gray, light-gray, blue, green, black, white, shiny-green, shiny-red

## Background Colors

red-background, dhw-gray-background, dhw-light-gray-background, white-background, blue-background, light-green-background, green-background, yellow-background

### Example

```
:dhw-red: Red Text.
```

### Fine-grained Control (Try to avoid!)

margin-none, margin-0-5em, margin-1em, margin-top-1em, margin-top-2em, margin-bottom-1em, margin-bottom-2em, margin-right-1em, margin-left-1em, padding-none, padding-0-5em, padding-1em, padding-top-1em, padding-top-2em

### Hiding slides (⚠ rst2ld only)

Use `hide-slide` to exempt it from slide generation.

### Example

```
.. class:: hide-slide
```

<Hidden Slide >

## Configuration

LectureDoc meta information:

`id` The unique identifier for the slide set. Required to store the current state of the presentation.

`slide-dimensions` the slides dimension (default: "1920x1200").

`first-slide` Determines the first slide that is shown (e.g., <Slide Number> or "last-viewed").

### Example

```
.. meta::
  :id: <unique id>
  :slide-dimensions: 2560x1440
  :first-slide: last-viewed
```

## Cheat Sheets with LD<sup>2</sup>

A cheat-sheet is a slide with the class `cheat-sheet-8-columns`.

### Template

```
.. class:: cheat-sheet-8-columns

<Title>
-----

.. container:: cheat-sheet-block

  .. rubric:: <TOPIC>

  .. rubric:: <SUB-TOPIC>
```

## Useful Role and Substitution Definitions

### Template

```
.. role:: incremental
.. role:: eng
.. role:: ger
.. role:: peripheral
.. role:: obsolete
.. role:: red
.. role:: gray
```

```
.. role:: light-gray
.. role:: blue
.. role:: green
.. role:: orange
.. role:: shiny-green
.. role:: shiny-red
.. role:: dark-red
.. role:: black

.. role:: raw-html(raw)
:format: html
```

## Links

[DocUtils \(rst reStructuredText\)](#)

[Example Slide Sets](#)

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### Template

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```